



### 8. Location of Sampling Units:

Accurate location is necessary for getting true results. The content of validity must be there in data that depend upon sample representative.

### 9. Combat with Field Problems:

Problems are the part of field research, which should be anticipated and overcome by researcher. Few problems are mentioned below:

- (i) Without establishing rapport, respondents always rise questions to interviewers like "who are you"?
- (ii) Gathering or over crowding, around respondent, while conducting interview.
- (iii) Getting of negative response from respondents.
- (iv) Language problem.

### 10. Supervision of Field Work:

Supervisor must be experienced and full time employee. He must have clear understanding of research with leadership qualities. He should organise groups, daily reports, output quotas for interviewers. He must be task oriented person and should provide all resources and facilities to his interviewers.

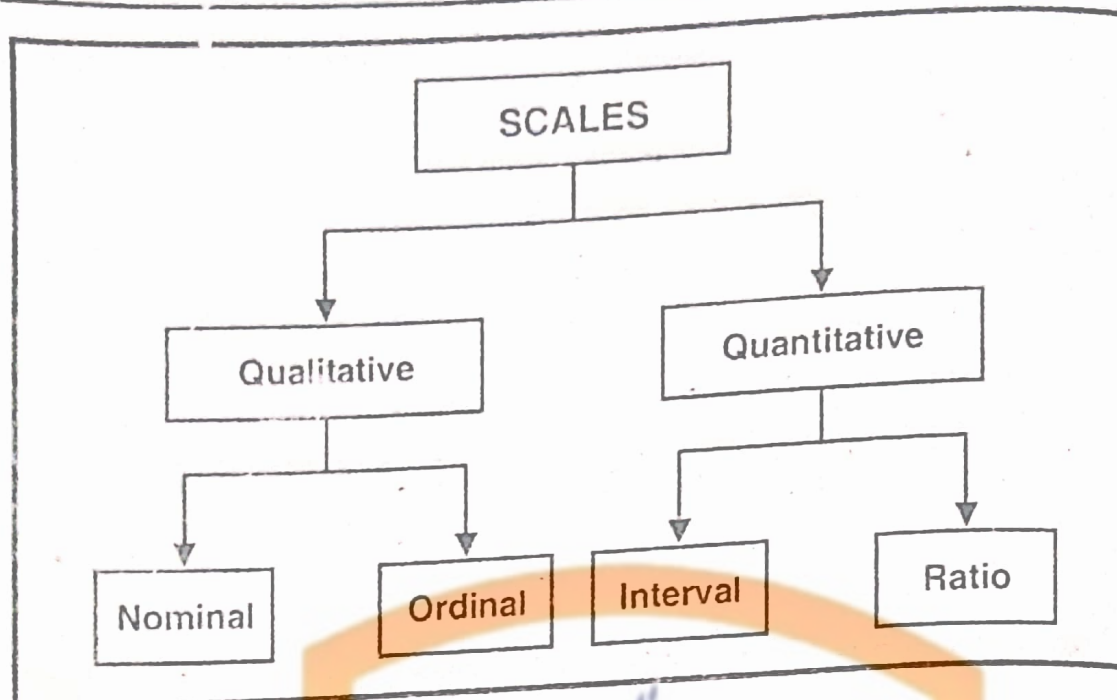
### Conclusion:

This method is popular in Pakistan for getting primary data. Personal interviews, telephonic interview or mail questionnaires are common methods of communication in survey.

It helps us to get information, statistics with the help of questionnaires to understand and overcome the socio-economic problems.

## DIFFERENCE BETWEEN SURVEY INTERVIEW AND FIELD INTERVIEW

| Typical Survey Interview                                                         | Typical Field Interview                                                                               |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1. It has a clear beginning and end.                                             | 1. The beginning and end are not clear. The interview can be picked up later.                         |
| 2. The same standard questions are asked of all respondent in the same sequence. | 2. The questions and the order in which they are asked are tailored to specific people and situation. |



## CLASSIFICATION OR TYPES OF SCALES

Qualitative scales study the qualitative phenomena like, age, religion, attitude, opinion, diseases etc.

Qualitative scale has two types:

1. Nominal scales
2. Ordinal scales

### 1. Nominal Scales:

For classifying or separating the properties into different classes, nominal scale is being used. In this scale arbitrary numbers are assigned to different categories of observations. Nominal scales are used in classification and identification. The categories scale must be arranged in manner that depicts each measurement in to one category.

For example, classifications according to religion, such as Islam, Christianity, Hinduism, (1, 2, 3). Gender is assigned number as male or female (1, 2). Blood group is recorded as A<sup>+</sup>, B<sup>+</sup>, O<sup>+</sup>, A<sup>-</sup>, B<sup>-</sup> (1, 2, 3, 4, 5).

Nominal scales are used in economics to study the consumption of different classes, middle-income group, and higher income group. In other social sciences, it is studied for religion, race, occupations and other behavior etc.

Another approach is used to measure the homogeneity is Cronbach's Co-efficient alpha and second is Kuder-Richardson formula 20.

### Common Used Scales:

In our daily life we use many measure. For example before we drive the car we check the gas gauge to be sure I can make to university. We check our speedometer while driving on motorway, in extreme hot and cold weather we check the temperature often, we also measure the nonphysical subjects in daily life, e.g. We say the food at this restaurant is very excellent, Sam is pretty girl, Ahmed is very smart guy, Samreen always has a negative attitude towards life.

### Index and Scales:

Index and scales are interchangeably and often confuse. One researcher scale is another index and scale also measure ordinal and interval measures of variables.

Both scale and indexes helps the researcher about the information of variables and make the quality of measurement possible scales and index increases the reliability and validity.

### Commonly used Scales:

#### Likert Scale:

Renis Likert developed this scale in 1930. To provide an ordinal-level measures of a person's attitude in Likert scale the score on the scale is computed by summing the number of responses the person gives.

Likert scales measure the degree of any qualitative thing is requires to measure. We can use in Likert scale to measure the degree of agreement and disagreement on a particular statement.

Likert scale has a minimum of two categories such as agree and disagree but it is usually better to use four to eight categories. It is all up to researcher that how he combine or collapse categories after the data collection.

The categories can be increase at the end by adding

“Strongly agree”

“Some what agree”

“agree”

Example of Likert scale problem “Standards of Education”.

These are arbitrarily numbers assigned to degrees. It can also be used like.

|                |       |            |          |                   |
|----------------|-------|------------|----------|-------------------|
| Strongly agree | Agree | No/opinion | Disagree | Strongly disagree |
| -2             | -1    | 0          | 1        | 2                 |

### Example of Using the Likert Scale to Create Indexes

Sinderman and Hagen (1985) created indexes to measure beliefs about equal opportunity for woman index for equal opportunity for women.

1. Women have less opportunity than men to get the education for top jobs.

|                |                 |                    |                     |            |
|----------------|-----------------|--------------------|---------------------|------------|
| Strongly Agree | Some what Agree | Some what disagree | Disagree great deal | Don't know |
|----------------|-----------------|--------------------|---------------------|------------|

2. Many qualified women cannot get jobs; men with the same skills have less trouble

|                |                 |                    |                     |            |
|----------------|-----------------|--------------------|---------------------|------------|
| Strongly Agree | Some what Agree | Some what disagree | Disagree great deal | Don't know |
|----------------|-----------------|--------------------|---------------------|------------|

3. Our society discriminates against women.

|                |                 |                    |                     |            |
|----------------|-----------------|--------------------|---------------------|------------|
| Strongly Agree | Some what Agree | Some what disagree | Disagree great deal | Don't know |
|----------------|-----------------|--------------------|---------------------|------------|

Scoring: for all items

Strongly agree = 1,                      Some what agree = 2

Some what disagree = 4,              Disagree a great deal = 5

Don't know = 3

4. Disagree a great deal = 5, don't know = 3

Highest possible index score = 15, respondent feels opportunities for woman are equal lowest [possible Index score = 3, respondent feels opportunities are not equal.

*Source: By Newman (Social Research Method)*

### Semantic Scaling:

The semantic differential is a series of attitude scales. This popular attitude-measurement technique consists of presenting an identification of a company, measurement technique consists of presenting an identification

income than the similar cloth merchants running their business in Ichhra Market and Anarkali. Therefore results from only covering Liberty Market and Pace and ignoring Ichhra and Anarkali will produce error.

### Precision:

When precision is estimated by the standard error of estimate, a type of standard deviation measurement. As much as standard error will small, precision will be higher. Good sample always has a small standard error or estimates.

But it is also a fact that all sample does not provide precision. small size sample can produce different standard error than large.

In fact no sample always fully represent its population. The numerical value from population may be differ from sample numerical values, this is called sampling error.

### How Sample will be Select:

In selecting sample following things are taken into account, each has unique information. In selection procedure we shell consider the followings:

1. What is relevant population?
2. What are the parameters of interest?
3. What is a sampling frame?
4. What is the type of sample?
5. What is a sample size?
6. How much will it cost?

## TYPE OF SAMPLING

There are basically two types of sampling:

1. Probability Sampling
2. Non-Probability Sampling

### 1. Probability Sampling:

A simplest form of probability sampling is un-restricted simple random sample, stratified sample, cluster sample, double sample.

Probability sampling requires four considerations:

- Develop the methods for selecting sample.
- Determine the sample size.
- Always address to non-response population.

### Simple Random Sampling:

In simple random sampling, there is an equal probability of each population and each sample of being selected. In simple random sampling, researcher is always interested to increase efficiency in sampling. This sampling technique has a quality to increase efficiency. Researcher uses the following technique to increase efficiency.

1. Holding accuracy and decreasing cost.
2. Holding cost constant and increasing accuracy.
3. Rate of increasing the accuracy should be faster than the rate of cost.
4. Rate of decreasing the accuracy should be at a slower rate than the rate of cost.

**Note:** Sampling efficiency is the ratio of accuracy over cost.

It is very convenient process of sampling. It has an equal chance of probability of selecting the elements of population e.g.

If we want analyze the selection of teachers, what are necessary grounds.

We want to study this project from survey of different colleges. Then there is chance of selecting all the students from same college or there is chance that every college has its representation equally.

Some experts opinion is to choose 10 percents of population when sample size is small.

According to Ralph L. Rosnow:

*Random Sampling is a sample chosen by chance procedure and with known probabilities of selection so that every individual in the population will have the same likelihood of being selected.*

### Stratified Sampling:

Stratified sampling has a quality to increase the sampling efficiency by increasing the accuracy at faster rate than the cost. Both accuracy and cost increase. Due to variables used to form groups stratification is almost efficient statistically than simple random sampling and at worst is equal to it. It is used where target is known the features of certain subgroups of population it is also useful when different methods of observation are under consideration.

| STRATIFIED SAMPLING                                                                                                                     | CLUSTER SAMPLING                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 1. Population is divided into sub-groups. Each has the same representation in sample. But these are selected under particular criteria. | 1. We make sub-groups of population, with a few element in it. It has also some criteria to select sample. |
| 2. We try to create heterogeneity and homogeneity within sub-groups.                                                                    | 2. We try to create heterogeneity and homogeneity within sub-groups but we usually get the reverse.        |
| 3. We randomly choose elements from each sub-groups.                                                                                    | 3. We randomly chose a number of the subgroups and then study in depth.                                    |

Cluster sampling has a little difference from stratified sampling. It is also divided into subgroups, but it does not represent the population fully. e.g.

In election one party wins more numbers of seats of National Assembly but over all gets less number of votes from whole country in total than second party who gets more numbers of votes of public but gets less numbers of seats.

### Systematic Sampling:

It has equal chance of selecting of all the population in sample. But it requires proper formula or technique or planning. In this type researcher remains impartial but for researcher it is necessary to have complete list of all the elements of population. e.g.,

From the strength of 50 students in a class if we want to select 5 students then the students are divided into 5 groups. It means list is distributed of every 10 students and then making the list of first 10 students then next students and so on. Finally from these five lists of consist of 10 students, draw is made of one student from each list. Then there is equal chance of students to be selecting for sample from five lists.

### Complex Probability:

An efficient sample is one, which provides greater precision, means low error of the mean with a smaller sample size.

# CHAPTER

# 1

## RESEARCH METHODS: AN INTRODUCTION

### Introduction:

Research means, seeking the facts or realities, or analytical research, scrutinize the problems.

Infact research is purulent of two words "re" and "search", which means to search again. It is in the nature of human being to keep himself or herself active in seeking the facts.

*"Research is a systematic and refined technique of thinking employing specialised tools, instruments and procedures in order to obtain more adequate solution of a problem than would be possible under ordinary means. It start with a problems, collects data or facts, analysis them critically and reaches decisions based on the actual evidence".*

Another modern and broad concept is to provide meaningful investigation into any field of academic.

Before we procecd further let us know what Research is?

*"A research is a careful study or investigation of existing facts in order to discover new realities."*

*"Research is sufficiently objective and systematic to make possible classification, generalization and verification of the data observed".*

**Lundberg.**

*It is essentially a systematic inquiry seeking facts through objective verifiable methods in order to discover the relationship among them, and to deduce from them broad principles or laws".*

**Gopal**

*Research is a systematic and refined technique of thinking, employing specialized tools, instruments and procedures in order to obtain more adequate solution of a problem than would be possible means. It starts with a problem, collects data or facts, analyses them critically and reaches decision based on the actual evidence".*

**Crawford**

## **TYPES OF RESEARCH**

Basically there are two ends of research:

- (i) One is to solve current problems in work setting, or any micro and macro economics problem.
- (ii) Next one can contribute knowledge in a particular area of research e.g. if you want to analysis the causes of inflation. Regarding this problem if research improves your knowledge.

**However generally there are two types of research known as:**

- (i) Basic research
- (ii) Applied Research

### **Basic Research or Theoretical Research:**

Basic research deals with natural things and explains it in careful examination. Infact this research expands the limits of knowledge by establishing principles laws and things. This research provides foundation for further study.

In short we can say to discuss immediate problem, not to only solve it but more than to provide more knowledge and understanding the phenomena that occurs and to build theories based on the research outcomes. This research develops new scientific ideas and new ways of thinking about the problems.

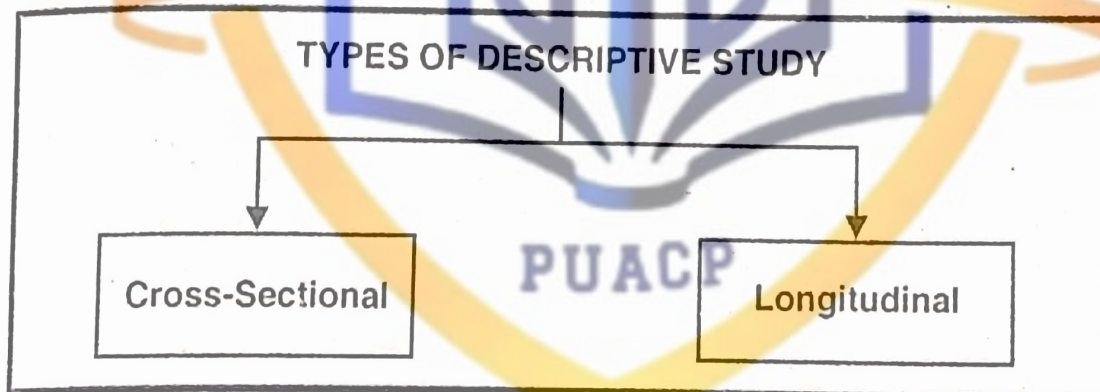
If an economist wants to know about rural and urban poverty levels goes into field and collects information through questionnaire from various households about their consumption.

To describe the characteristics of variables we use descriptive research it describes the parameter of system, or economic phenomena, market phenomena, and marketing prediction. For example:

1. A profile of various indices was prepared by United Nation regarding, health, education and clean water available in Pakistan etc. to compare with other developing countries indices.
2. Pre-capita income of Pakistan measure as indices to know poverty, compare with per capita income of other developing countries as indices to know poverty.
3. In marketing descriptive studies determine buyer perceptions of product characteristics and audience profiles for media such as television and magazines.

Descriptive research design can utilize one or more of following sources of data.

1. Primary Data
2. Secondary Data



### Cross-Sectional:

A cross sectional study provides information about the situation that exists at one point of time or a study or in which various segment of population are sampled at a single point in time is called cross-sectional study. A nationwide survey was taken to examine the different attitudes of cross-sections of the public toward the arts, and one aspect of the survey dealt with museums. In general, the public's attitudes toward museums were very positive. Museum preferences varied by demographics or cross-

sections of the population. People in towns and rural areas showed greater interest in historical museums, whereas city and suburban residents leaned more heavily toward art museums. The young (16- to 20-year-olds) were more interested than others in art museums and less interested in historical museums.

This study was a cross-sectional study because the data were collected at single point in time. In such a study, various segments of the population are sampled so that relationships among variables may be investigated by cross-tabulation. Most surveys fall into this category.

The typical means of analyzing a cross-sectional survey is to divide the sample into appropriate subgroups. For example, if the HEC (Higher Education Commission) in Pakistan expects job title or position to influence attitudes toward continuing-education programs, then its survey data are broken down into sub-groups based on job title and analyzed to see if there are any similarities or differences among the subgroups.

(Adapted from William G. Zikmund; Business Research Methods)

### **Longitudinal Study:**

A survey of respondents at different points in time, thus allowing analysis of response continuity and changes over time is called longitudinal study. In a longitudinal study respondents are questioned at different moments in time. The purpose of longitudinal studies is to examine continuity of response and to observe changes that occur over time. For example, in 2010, 59 percent of respondents ranked college teacher as an occupation high or very high in honesty and ethics. Longitudinal studies of this type are sometimes called cohort studies, because similar groups of people (cohorts) are expected to be in each sample. This research is conducted by many syndicated polling services, Gallup Poll services etc. For example, in election studies that monitor voters' attitudes toward issues or their awareness of a candidate when a new political campaign is being launched frequently use successive samples. Having two or more sample groups avoids the problem of response bias that might result from a prior interview. A respondent who had been interviewed in an earlier survey about a certain issue or candidate may become more aware of the issue or pay more attention to the candidate after being interviewed. Using different samples eliminates this problem. Of course, researchers can never be sure whether changes in the variable being measured result from having different people in the sample or reflect an actual change in the variable over time.

(Adapted from William G. Zikmund; Business Research Methods)

is necessary to know and use a wide variety of bibliographies. It includes the following:

1. National Bibliography
2. Trade Bibliography
3. Historical Bibliography
4. Research Bibliography
5. Subjective Bibliography
6. References and Bibliography
7. Bookish Bibliography

(Note: Students can see detail in Appendix)

### 1. National Bibliography

National bibliography represents the nation. It contains all kind of documentary publication, trade or non-trade material includes.

e.g. First Magazine of America (Trade Related) is continuous publishing since 1872.

### 2. Trade Bibliography

It contains research thesis or dissertations list. A famous bibliography is publishing from Mishigon since 1938.

### 3. Historical Bibliography

When reference is related with a specific subject it is called subjective bibliography. E.g., Shakespeare bibliography.

### 4. Research Bibliography

It contains the list of topics, references, and formulas, numbers that are in systematic way. e.g. Magazines and Newspaper use it frequently.

## REFERENCES AND BIBLIOGRAPHY

### Format for Referencing the Articles Relevant to the Study:

A bibliography is the listing of the work that is relevant to the main topic of research interest arranged in the alphabetical order of the last names of the authors. A reference list is a subset of the bibliography, which includes details of all the citations used in the literature survey and elsewhere in the paper, arranged again in the alphabetical order of the last names of the authors. These listings enable the reader to access the original works cited, if there is a need to do so.

## 2. Parameters of Interest:

Population parameters mean what are descriptions (For example, incidence proportion, mean, variance) of variable in the population.

Actually sample statistics are only used to study or estimate the population parameters.

These parameters of interest are different for different types of data.

|                                                         |          |                                                                             |
|---------------------------------------------------------|----------|-----------------------------------------------------------------------------|
| 1. Attendance at a Economic Seminar.                    | Nominal  | Participation (yes, no)                                                     |
| 2. To analyze the performance of teacher in college.    | Ordinal  | Cooperation with students, strong, medium, low or always, sometimes, never. |
| 3. Mean temperature of Murree.                          | Interval | Temperature in degree                                                       |
| 4. An average number of visits per month to pace store. | Ratio    | Actual number of visitors at pace.                                          |

Researcher always uses interval and ratio scales. When they use the sample to estimate the population mean and the sample standard deviation to estimate the population standard deviation.

In case of nominal or ordinal scales, researcher uses the sample proportion of incidence to estimate the population proportion and the  $p_q$  to estimate the population variance. Proportion is very much necessary for nominal data and often used for other measures as well.

## 3. Sampling Frame:

The list of population members or elements is required to draw sample. It is called sampling frame. The list of population's members should be corrected and complete. TRG call center require keeping record of all the calls of their clients about their products in order to study the growth sale of product.

Researcher often has to face irrelevant members of population in sample frame.

Sometimes researcher used telephone directory to draw a sample of business telephone numbers. Screening procedure is helpful to exclude the non-members of population for his or her area of interest.

### How to Stratify?

Stratification decision depends upon data availability. But there are three kinds of decisions.

1. What kind of stratification base to be used?
2. How many strata should be used?
3. What should be strata sample sizes?

The perfect stratification would be based on the principal variable of study. For example if we want to know that how many times per week students would use dining club. Then on expected number of occasions we stratification would be made. Sometime, we pick a variable for stratifying which we are sure has correlation with the frequency of club use per month.

### Cluster Sampling:

In cluster sample, the sampling efficiency can be improved by decreasing cost at a faster rate than accuracy. Cluster also divide the population into sub-groups, however in cluster a random sample of subgroups is selected. In cluster sampling all the members are interviewed in sub-groups. Cluster sample also reduces the cost but with disadvantage of being formed heterogeneous cluster.

### Cluster Efficiency:

Statistical efficiency in cluster sampling is less than simple random sampling. Cluster has a quality of homogeneous. e.g., All families living in "DHA" (a typical cluster) will be almost similar in social class and income level.

### Cluster Design:

In cluster design, many interrelated questions can be asked. While they are discussed separately:

1. How cluster are the homogeneous depends upon many internal characteristics. By increasing intra-cluster variance, we can improve the efficiency by constructing clusters. In area sampling we can combine contiguous blocks of different income groups or social classes.

## OPEN-ENDED OR OPEN RESPONSE QUESTIONS

In open ended questions, respondents are allowed to answer in any way they like and gives freedom to respondent. Respondent always appreciate his freedom, if given. People usually write their marginal comments in mail surveys particularly in a situation, where they feel that there is no category for adequate response. The right decision of question designing depends on degree and form of structure imposed on responses. The design options are in the range from open response questionnaire to closed response questionnaire.

### Situational Determinants:

Khan and Cannell pointed out five factors to take decision regarding choice of question format. In both open and close-ended questionnaire, these five factors are considered.

1. Interview objective.
2. Level and understanding of respondent.
3. Degree respondent has thought through the topic.
4. Make easy environment for communication with respondent.
5. Degree to which the above respondent factor.

### Advantages of Open-ended Questionnaire:

1. Open-ended questionnaire is used where the objective is to get the opinions and degree of knowledge from the respondent.
2. It is also used where objective is to get, suggestions, dates, events, particularly when probes are used.
3. It also helps to determine the degree or intensity about certain feelings and expressions.
4. This is also suggested to use where respondent has any ambiguous idea or information. These kind of situations often comes in explanatory research. E.g., Examples of open-ended questions when respondent is asked that:

Q.1: "State five things that are interesting and challenging in the job.

Q.2: What respondents like about their supervisors or their work environment.

**CLOSED-ENDED QUESTIONS**

Closed-ended questionnaire is used to classify the respondent on some point of view. There is no open options for respondent e.g. If the objective of a study to determine whether respondent like certain corporate policy or not. So the scope of respondent's opinion here is limited.

**Advantages of Closed-Ended Question:**

1. If respondent has already developed a clear opinion on the topic, then it better justified to use closed-ended questionnaire.
2. It takes less motivation among respondents.
3. Respondent in answering questions feel less threat as compared to open-ended questionnaire.
4. Closed response is advocated where there is clear frame of reference.
5. It is used where researcher feels respondent's level of information is predictable.
6. It requires less skills of interviewer and much easier to code and analyze.
7. After adequate information and measures it will perform as effectively as open-ended questions.
8. It is superior or equal to open-ended questions in case of experimental studies.

Examples of closed-ended questions it has two way or multiple choice response but choices are given always. E.g.,

Q.1: It must be paid attractive package to teachers to increase the education standard.

- (i) Strongly agree
- (ii) Agree
- (iii) No opinion
- (iv) Disagree
- (v) Strongly disagree

Q.2: Women have less opportunity than men to get high grade jobs.

- (i) Strongly agree

(e) **Measurement Error:**

Measurement error occurs because of variation between the information sought by a researcher for a study and information taken by particular procedure used for measurement. For example if researcher wants to know individual income (P.M) of a respondent while question developed as an annual individual household income, then this is a measurement error.

(f) **Experimental Error:**

An experimental error is due to experimental designed of any cause-effect relationship between two variables. This always occurs due to improper designed.

(g) **Data Analysis Error:**

Data analysis means carefully coding of questionnaires, otherwise, coding, editing and interpretation if not property done, data analysis error may occur.

(ii) **Administering Error:**

An error due to administration of survey instrument is administering error. This happened by mistake during administering the questionnaire.

There are three factors of this cause:

- (a) Questioning error
- (b) Recording error
- (c) Inference error

(a) **Questioning Error:**

This error is due to saying or addressing questions to the respondents. If the question is not asked as it is designed then this error occur.

(b) **Recording Error:**

This error arises due to improper recording of the respondent's response, if the interviewer does not hear property or accurately, it induces recording error.

(c) **Interference Error:**

In collecting data, if interviewer fails to follow exact procedure. For example if any time researcher fabricates the responses to a survey, it makes interference error.

Another approach is used to measure the homogeneity is Cronbach's Co-efficient alpha and second is Kuder-Richardson formula 20.

### Common Used Scales:

In our daily life we use many measure. For example before we drive the car we check the gas gauge to be sure I can make to university. We check our speedometer while driving on motorway, in extreme hot and cold weather we check the temperature often, we also measure the nonphysical subjects in daily life.e.g. We say the food at this restaurant is very excellent sam is pretty girl. Ahmed is very smart guy. Samreen always has a negative attitude towards life.

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The categories can be increase at the end by adding

"Strongly agree"

"Some what agree"

"agree"

Example of Likert scale problem "Standards of Education".

## METHODS OR BASIC STEPS FOR CONDUCTING SURVEY

There are some essential steps or methods:

1. Determine objectives
2. Methodology for research
3. Questionnaire preparation
4. Pre-test questionnaire
5. Selection of sample size
6. Select the field workers to collect data
7. Train field workers
8. Location of sampling units
9. Anticipate and overcome the field problems
10. Supervision of field work

### 1. **Determine Objective of Survey:**

Researcher should arrange meetings and discussions with senior executives of the sponsoring group. Discussion must be about plant tenure, needs, resources and time.

A work should be done with entire mutual consensus between sponsoring group and survey team.

### 2. **Decide the Research Methodology:**

Very clear design will direct the research team to follow his plan for achieving his objectives of research study.

### 3. **Questionnaire Preparation:**

Questionnaire preparation is not a simple technique, but it must contains the designed information require in the context of data.

The questionnaire should be in easy language, and avoid confusing and time consuming questions that are difficult to answer.

In Pakistan I personally advise to prepare it in Urdu language.

### 4. **Pre-test the Questionnaire:**

To check the adequacy of the questionnaire, this technique is followed before starting the field work to collect data.

Following are the main steps in the writing research proposal

1. Title and Title Page
2. Introduction
3. Statement of problem
4. Review of Literature
5. Objectives
6. Data sources
7. Methodology
8. Hypothesis
9. Model
10. Reference

Now will briefly discuss the following steps.

A title page of synopsis or research proposal should include title of research project, name of research (with qualification) name of supervisor place of work, and date (month and year) of submission e.g.

1. **Title and Title Page:**

**Title: Impact of Foreign Investment on Poverty Reduction**

From (1973 to 2005)

Submitted by: X.Y.Z

(M.A Economics) M.B.A

Supervisor: X.Y.Z

Place of work : X.Y.Z

(Institution name)

*Note:* Students must remember the topic for research should be selected carefully. It should be specific and well worded to reflect the nature of work involved as far as possible.

## 2. Introduction

It should provide a brief description to introduce the area of the proposed research work

## 3. Statement of Problem

Students should analysis the problem statement before selection of research area that it is suitable for research or not if it is suitable then what are its advantages e.g. in our study either foreign investment is specific sector can reduce poverty or nor.

## 4. Review of Literature

A review of the relevant literature showing the work done previously in the area of proposed research is essential to plan further research effectively. The information given in the review should be supported by references

## 5. Objectives

Broad objectives as visualized to achieve should be clearly outlined and these should be itemized. These objectives will indicate the major aspects of the study to be undertaken.

*Note:* Students must not that objective can be part of introduction as well.

## 6. Data Sources

Students should clearly mention the sources of data e.g.

“Economic survey of Pakistan”

FBS (Federal Bureau of Statistics) web site.

From (1973 to 2005)

## 7. Hypothesis

$H_0$ : Foreign direct investment can reduce poverty.

$H_1$ : FDI does not has impact on poverty reduction

## 8. Model

In practice, the research model is developed on the basis of dependent, independent and dummy variables for the thesis of, MA, M.Phil and Ph.D.

*Note:* In exam students will develop Hypothetical model on the basis of the variables that they have introduced in research with the help of econometric techniques.

### References and Bibliography

Synopsis or research proposal should contain at the not a list of references. These should be written following a standard pattern.

- (i) Writers alphabetically name of reference
- (ii) Sir name should write First.
- (iii) Year of work
- (iv) Title of work

*Note:* Detail see in Chapter Bibliography.

### 10. Plan of Work and Methodology

A plan of work describing the various aspects of the study in a logical sequence along with the methodological to be employed are the most important aspects of any research plan. Sufficient details to demonstrate that the researcher has a fairly good idea about the nature of work likely to be involved should be provided. In case of experimental sciences, it should be indicated which equipment and experimental procedures will be used to obtain the result. In case of social sciences what resources, materials will be used whether the required information will be obtained from primary or secondary sources, etc.

Journal name (if it is from journal)

e.g. Gergoroi and Jong Whalee (1988)

"How does foreign direct investment effect economic growth".  
Journal of international economics pp 127-135

## HYPOTHETICAL RESEARCH PROPOSALS

### RESEARCH PROPOSAL-I

#### Macro Economic Policies and its Impact on Inflation

*Note:* Student must write the title brief and comprehensive in meaning that must relate their research work and tile page is already mentioned.

#### 1. Introduction:

Macroeconomic policies play a critical role in determining the rate of inflation. For example growth of money supply, tax, fiscal policy, exchange rate depreciation and increase in demand for domestic goods are the factor of cause of inflation.

For example, a teacher conducts test on weekly basis in class before giving them textbook. Then after he gives this textbook to students for study purpose and takes again test on weekly basis. In checking scores if he feels improvement then this is time series design.

Basic time series design

01 02 03 04 05 X 06 07 08 09 010

1. From 0 to 5 shows pre test
2. X means treatment
3. Zeros of 6 to 10 means post test

### Conclusion:

These three process are very much common in research. It is difficult to distinguish among all these, because of overlapping characteristics in the concepts. These terms have been defined and explained differently by different researcher. All three have great importance but it requires specific steps in methodology. These are always enlisted in sequence.

## RESEARCH DESIGN

Three steps are to be noted in research design.

1. What is a research design?
2. Basic issue of a research design
3. Component of a research

### 1. What is a Research Design?

Research design is always necessary before the research is conducted.

A research design is a basic plan that guides the type of information to be collected, the source of data, and phase of analysis of the research project. A good design will make sure that information collected is consistent with the study objective and data are collected in a proper way.

### Good Research Design:

Every design has its own merits and demerits but a good research design should satisfy the following four conditions.

#### (i) Objectivity:

This relates to plan and interpretation of the results framed in such a way so that it is not influenced by the investigator or to the response.

#### (ii) Validity:

Any measuring instrument should measure only intelligence and not other factors. Good and minimizes errors. Hypotheses of cause-effect relationship. Particularly the role of confounding factors are threats to it. There are many other conditions. While in the field or extraneous factor is possible to attain what is actual conditions.

#### (iii) Reliability:

Reliability refers to the consistency of the results. This is very important through many ways. Simultaneously use of words reliability means repeated under same conditions give inconsistency. My weight measuring process again and again. Clothing, drinking, consistency is a result.

- Stability
- Reproducibility
- Equivalence

#### (iv) Generalizability:

The plan should be such that the results of the study are possible if it is applied to other situations.

|          |                                        |                                                        |                                                                                                            |
|----------|----------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Interval | Equality of differences<br>(2-1 = 7-6) | Attitudes<br>Opinions, Index<br>numbers                | Range Mean,<br>Standard<br>deviation,<br>Product-Moment<br>correlation, Factor<br>analysis and<br>ANOVA    |
| Ratio    | Equality of ratios<br>(2/4 = 4/8)      | Ages, costs<br>number of<br>customers sales<br>(units) | Geometric mean<br>Harmonic mean<br>Co-efficient of<br>variation and all<br>types of statistical<br>Methods |

Source: Uma Sekran (*Research Methods for Business*)

## TESTING THE GOODNESS OF MEASURE / SOUND MEASURE

Scales do not show always perfect or 100 percent accurate results. However, better use of instrument will ensure more accuracy of results. To develop reasonably good result, two criteria for goodness of measurement is required.

1. Validity
2. Reliability

### 1. Validity

A process, which is free of systematic and random error is called greater extent of validity.

Before we discuss validity, we shall focus on error. The error can affect the quality in various stages of research process. The error can be defined as the difference between the true mean value and observed mean value obtained from our research study.

The total error is consist of two components.

Total error = Sampling error + Non-sampling error

### Sampling Error:

Sampling error is the difference between a measure from representative sample and true measure from entire population. This means no sample can be perfect representative sample and thus sampling error is always there.

**Feel for Data:**

By checking central tendency, and the dispersion of the data, feel for data can attain. Researcher can use, mean, range, standard deviation, and variance to obtain feel for data, which shows how good the items and how they are measure. Otherwise if responses have a little variability, the researcher would think and suspect the questions were not well worded.

The mean, standard deviation, skewness, range can be obtained on computer through SPSS or different programmes.

For demographic analysis, researcher can also use, histogram, bar charts, ogive polygon by appropriately programming the computer.

The inter-correlation matrix of variables, irrespective of the hypothesis is directly related to this analysis or not. These all statistics test gives the researcher feel for data. To check the variables, whether they or closely related or unrelated the correlation study can also be taken into analysis.

If correlation between two variables is over .80, we might think, whether they are measuring same concept or different.

**TESTING THE GOODNESS OF MEASURES**

The scales developed are not 100 per cent and there are always error in measurements. However the use of better instruments will ensure more accurate results. Hence we need to assess the goodness of measures. There are two criteria for testing the goodness of measures:

- (1) Validity
- (2) Reliability

**Validity:**

The validity of a measure refers to the extent to which the measurement process is free from measurement error (i.e. systematic) and random errors.

**Measurement Errors**

Measurement error is the lack of correspondence between the number system and the phenomena being measured. It consists of two components: systematic error and random error.

Systemic error and random error together make up the total error.

$$\text{Total error} = \text{Systematic error} + \text{Random error}$$

## CHAPTER

## 5

## RESEARCH DESIGN IN RESEARCH PROCESS

### Introduction:

Earlier we mentioned research techniques with limitations. Now we will consider many different types of research approaches and the one or ones that can be used for given problem. In this chapter we will study research design then we shall discuss classification of design and several design.

### WHAT IS RESEARCH DESIGN?

Research design is always necessary before conducting research.

A research design is a basic plan that guides the type of information to be collected, the source of data and phase of analysis of the research project.

The research design constitutes the blueprint for the collection, measurement, and analysis of data.

We can also say overall planning before doing investigation.

It involves all the subject matter in the form of steps, ingredient from developing hypothesis, to final results, and its presentation.

It is the framework or configuration of the relations among variables to investigate.

## COMPONENT OF A RESEARCH DESIGN (OR) RESEARCH STEPS IN RESEARCH DESIGN

Any investigator, before getting on to the job of planning a research design for a proposed study, should be clear about the following aspects:

- What to observe?
- Whom to observe?
- How to observe?
- Why to observe?
- How to record the observation?
- How to analyze the observation?
- What inferences can be drawn?

Once the investigator has sufficient understanding about the above-mentioned aspects. He/she can report to the job of planning a research design. In doing so he proceeds step by step in the following order:

### (i) Title of the Investigations or Research:

The title of any research project should be:

- As brief as possible.
- As precise as possible.
- Should be in generalized terms.

e.g., Impact of FDI (Foreign Direct Investment) on growth and employment shows the what is in study.

### (ii) Purpose of the Study:

A brief mention of the importance of the area of research in the present context of social life should be attempted highlighting the main purpose that prompted the investigator to take up the recent study. e.g., To see whether there is a relationship between FDI, growth and employment.

### (iii) Critical Study of the Previous Research or Literature Review:

Critical understanding of previous research is very much necessary for new research to make it useful, meaningful and advantageous. Researcher for doing this can follow the following guidelines: